

Assignment 2 - Part 3.2: The Receptive Field Experiment

Feature Extraction & Visual Interpretability

1 Introduction

This experiment visualizes how the “focus” of a CNN evolves across layers. We extract activation maps from the first and final convolutional layers to show the transition from low-level features (edges, textures) to high-level semantic patterns (object shapes). Analysis is performed on MNIST and Tiny-ImageNet-10 using the same SimpleCNN architecture from Part 3.1.

2 Methodology

We use PyTorch’s forward hook mechanism to capture intermediate activations from the first and final convolutional layers. Activation maps from multiple channels are averaged to produce representative heatmaps for visualization.

3 Results & Visualizations

3.1 MNIST

Figure 1 shows the activation maps for a sample MNIST digit. The first convolutional layer responds to edges and local patterns, while the final layer shows concentrated activation on the digit’s overall shape.

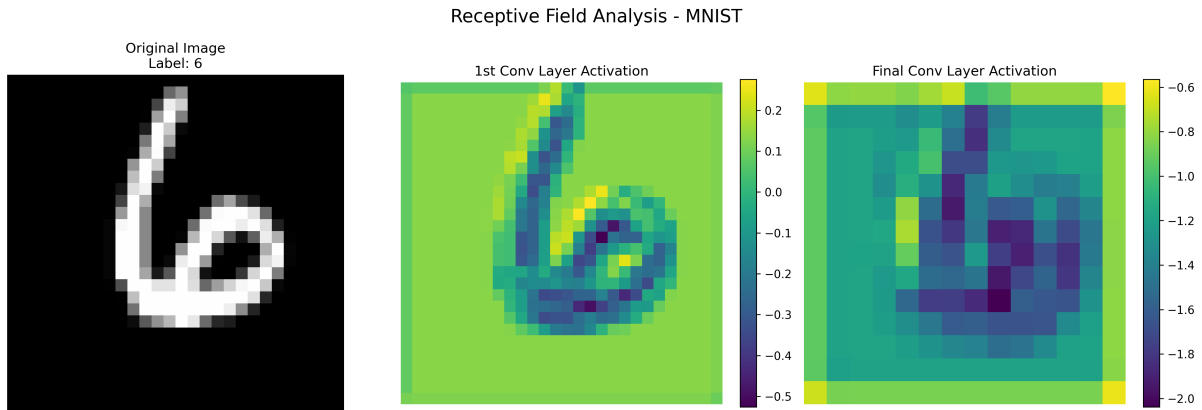


Figure 1: Receptive Field Analysis on MNIST. The first layer detects edges; the final layer captures digit-level features.

3.2 Tiny-ImageNet

Figure 2 presents activation maps for a Tiny-ImageNet sample. Early layers detect color blobs and chromatic edges, while deeper layers respond to object parts and semantic regions.

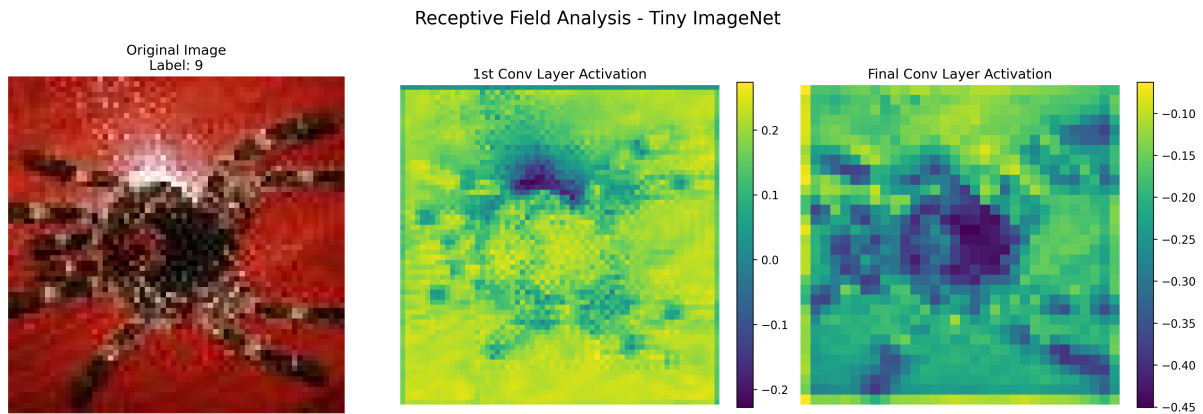


Figure 2: Receptive Field Analysis on Tiny-ImageNet. The first layer detects textures and color edges; the final layer focuses on object-relevant regions.

4 Analysis

How the Focus Changes:

- **1st Conv Layer:** Acts as low-level feature detector, responding to edges, corners, and textures. Small receptive field ($\approx 3 \times 3$).
- **Final Conv Layer:** Combines features into high-level representations. Larger receptive field captures global context and object-level patterns.
- **Receptive Field Growth:** Each layer “sees” more of the input, enabling the network to integrate spatial information hierarchically.

MNIST vs. Tiny-ImageNet:

- **MNIST:** Cleaner activation maps due to simple grayscale digits. Final layer focuses on digit shape.
- **Tiny-ImageNet:** Richer feature hierarchies. First layer learns chromatic features (color-opponent filters). Final layer activations are more distributed due to object variability and background clutter.

Key Observations: The visualization confirms that CNNs automatically learn hierarchical representations from simple to complex, mirroring biological visual systems. The same architecture adapts to different datasets, demonstrating the flexibility of CNNs.